

Q 1. Evaluate the following integrals

a) $\int \frac{dx}{1 + \sin^2 x}.$

b) $\int \frac{\sqrt{x}}{1 + x^3} dx.$

Q 2. Evaluate the following integrals

a) $\int e^\theta \sin \theta d\theta$

b) $\int \sin(\ln x) dx$

c) $\int \arcsin x dx.$ (**HW**)

Q 3. Show that

$$\int_a^b \left(\int_x^t f(t) dt \right) dx = \int_a^b (x-a)f(x) dx$$

Q 4. Evaluate the following integrals

a) $\int \cos^3 2x \sin^5 2x dx$

b) $\int_{\pi/2}^{3\pi/4} \sqrt{1 - \sin 2x} dx$

Q 5. Evaluate the following integrals

a) $\int_{-\pi/2}^{\pi/2} \cos \theta \cos 7\theta d\theta$

b) $\int \tan^3 \alpha d\alpha$